MTH 4300
Spring 2026

Programming Problem Set 1: Intro to C++ and Functions

Directions: Download the template files provided on Brightspace. Submit your source code for each problem individually via Gradescope, in `.cpp` format.

Submission Reminder: Submitting for feedback is optional. If you submit by the deadline and want feedback beyond the autograder, indicate that in your submission. You may resubmit unlimited times.
Important: Passing all test cases does not guarantee correctness. The autograder tests limited inputs. Test additional cases yourself.
AI Usage: If you use AI, document what you asked and what you learned in your code comments. State which tool (ChatGPT, Claude, Copilot, etc.). Be skeptical. AI can be confidently wrong. Your prompts should never include the entire problem; use AI to get unstuck, not to replace your thinking. If you collaborate with peers, acknowledge their contributions.
Using AI to Debug: If stuck, prompt AI to tutor you rather than solve for you:

> *"My program should [EXPECTED BEHAVIOR] but I'm getting [ACTUAL BEHAVIOR]. Don't explain the fix, ask me short guiding questions to help me find it myself. Keep a back-and-forth conversation. I should do most of the thinking."*

If AI gives too much detail: *"That's too much. Just hint at the next step."*
Note: While learning, avoid IDE tools that autocomplete or generate code for you (e.g., Copilot, IntelliSense suggestions). Writing code yourself builds the understanding you'll need for exams.

Be sure to read the SPECIFICATIONS carefully! And write comments!

Specifications and Notes (for both parts):

- You can only use the header `iostream`.
- You cannot use sequences/arrays, vectors, or any advanced data structure that can store an arbitrarily long collection of integers.
- No input prompts should be used, and your output should end with a new line.

Note: Each problem has test cases on Gradescope. All test cases are visible, so you can see exactly which cases your code passes or fails. Use this feedback to improve your solution. You may resubmit as many times as you like. Remember, these assignments are for practice; use them to prepare for exams!

To pass the test cases, the output should be *exactly* as specified in the samples below. When I say exact, I mean identical, down to the spacing. (There should be no additional outputs other than those specified.) In this class, I have written autograders for most assignments, which run on your code as soon as you submit it and let you know if you pass the tests I've written. Those tests check if your output matches what I've specified down to the spaces, so you must follow my specifications exactly. (This may seem annoying, but it is a valuable skill to master. It is very common in computer science to have precise instructions for something to work or for a client to be happy.) The benefit of autograders is that you can immediately know if your code isn't doing exactly the right thing and fix it before the deadline!

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1a) **distinctChars.cpp**

Write a program that reads three characters (i.e., `char`) from standard input. The program should then determine the number of *distinct* characters entered and output the result.

Below are some examples of input and the corresponding output.

Input: `a b 2`
Output: `3`

Input: `A a a`
Output: `2`

Input: `# # #`
Output: `1`

1b) **functionSum.cpp**

Suppose we are given the function

$$f(x, y) = x^4 + y^3 - 2xy.$$

Write a program that reads a positive integer n from standard input, followed by n pairs of integers $x_1, y_1, x_2, y_2, \ldots, x_n, y_n$. (For example, the input `1 3 4` would correspond to $n = 1$, $x_1 = 3$, and $y_1 = 4$.)

The program should then determine the result of the alternating sum

$$-f(x_1, y_1) + f(x_2, y_2) - f(x_3, y_3) + \cdots + (-1)^n f(x_n, y_n)$$

and then output the result.

Below are some examples of input and the corresponding output.

For $n = 2$, $x_1 = 1$, $y_1 = 1$, $x_2 = 1$, $y_2 = 2$:
Input: `2 1 1 1 2`
Output: `5`
Since $-f(1, 1) + f(1, 2) = -(1^4 + 1^3 - 2(1)(1)) + (1^4 + 2^3 - 2(1)(2)) = 5$.

For $n = 3$, $x_1 = -1$, $y_1 = 0$, $x_2 = 2$, $y_2 = -2$, $x_3 = 0$, $y_3 = 1$:
Input: `3 -1 0 2 -2 0 1`
Output: `14`
Since

$$\begin{aligned} -f(-1, 0) + f(2, -2) - f(0, 1) &= -((-1)^4 + 0^3 - 2(1)(0)) \\ &\quad + (2^4 + (-2)^3 - 2(2)(-2)) \\ &\quad - (0^4 + 1^3 - 2(0)(1)) \\ &= -1 + 16 - 1 \\ &= 14 \end{aligned}$$